Year 11 Foundation

REVISION PACK

Number 9

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 9 · Percentages of amounts (Lesson 6.1)

Q	Working	Answer
a percentage of an amount		
1.1	$15\% \text{ of } 80 = \frac{15}{100} \times 80$ $= 12$	**12 KD**
1.2	$65\% \text{ of } 360 = \frac{65}{100} \times 360$ $= 234$	**234 KD**
1.3	$12\% \text{ of } 650 = \frac{12}{100} \times 650$ $= 78$	**78 KD**
1.4	$25\% \text{ of } 180 = \frac{25}{100} \times 180$ $= 45$	**45 KD**
1.5	$5\% \text{ of } 340 = \frac{5}{100} \times 340$ $= 17$	**17 KD**
1.6	$75\% \text{ of } 96 = \frac{75}{100} \times 96$ $= 72$	**72 KD**
writing one number as a percentage of another		
1.7	$\frac{18}{25} \times 100$ $= 72\%$	**72%**
1.8	$\frac{39}{60} \times 100$ $= 65\%$	**65%**
1.9	$\frac{63}{84} \times 100$ $= 75\%$	**75%**

1.10	$\frac{27}{30} \times 100$ $= 90\%$	**90%**
1.11	$\frac{21}{35} \times 100$ $= 60\%$	**60%**
1.12	$\frac{56}{80} \times 100$ $= 70\%$	**70%**
a percentage of the rest		
1.13	$\text{rest} = 1000 \times 0.8 = 800$ $\text{rent} = 800 \times 0.3 = 240$	**240 KD**
1.14	$\text{rest} = 1500 \times 0.9 = 1350$ $\text{rent} = 1350 \times 0.4 = 540$	**540 KD**
1.15	$\text{rest} = 800 \times 0.75 = 600$ $\text{food} = 600 \times 0.5 = 300$	**300 KD**
1.16	$\text{rest} = 2000 \times 0.8 = 1600$ $\text{food} = 1600 \times 0.25 = 400$	**400 KD**
1.17	$\text{rest} = 1200 \times 0.75 = 900$ $\text{rent} = 900 \times 0.2 = 180$	**180 KD**
1.18	$\text{rest} = 900 \times 0.9 = 810$ $\text{food} = 810 \times 0.5 = 405$	**405 KD**

Number 9 · Percentage increase, decrease and profit (Lesson 6.2)

Q	Working	Answer
increasing an amount		
2.1	$\text{multiplier} = 1.1$ $40 \times 1.1 = 44$	**44 KD**
2.2	$\text{multiplier} = 1.14$ $75 \times 1.14 = 85.5$	**85.5 KD**

2.3	$\text{multiplier} = 1.08$ $250 \times 1.08 = 270$	**270 KD**
2.4	$\text{multiplier} = 1.05$ $120 \times 1.05 = 126$	**126 KD**
2.5	$\text{multiplier} = 1.25$ $64 \times 1.25 = 80$	**80 KD**
2.6	$\text{multiplier} = 1.3$ $90 \times 1.3 = 117$	**117 KD**
decreasing an amount		
2.7	$\text{multiplier} = 0.9$ $3000 \times 0.9 = 2700$	**2700 KD**
2.8	$\text{multiplier} = 0.85$ $7200 \times 0.85 = 6120$	**6120 KD**
2.9	$\text{multiplier} = 0.75$ $480 \times 0.75 = 360$	**360 KD**
2.10	$\text{multiplier} = 0.8$ $350 \times 0.8 = 280$	**280 KD**
2.11	$\text{multiplier} = 0.95$ $9000 \times 0.95 = 8550$	**8550 KD**
2.12	$\text{multiplier} = 0.85$ $640 \times 0.85 = 544$	**544 KD**
percentage profit		
2.13	$\text{profit} = 60 - 50 = 10$ $\frac{10}{50} \times 100 = 20\%$	**20%**
2.14	$\text{profit} = 46 - 40 = 6$ $\frac{6}{40} \times 100 = 15\%$	**15%**
2.15	$\text{profit} = 100 - 80 = 20$ $\frac{20}{80} \times 100 = 25\%$	**25%**

2.16	$\text{profit} = 78 - 60 = 18$ $\frac{18}{60} \times 100 = 30\%$	**30%**
2.17	$\text{profit} = 220 - 200 = 20$ $\frac{20}{200} \times 100 = 10\%$	**10%**
2.18	$\text{profit} = 20 - 16 = 4$ $\frac{4}{16} \times 100 = 25\%$	**25%**

Number 9 · Reverse percentages (Lesson 6.4)

Q	Working	Answer
price before an increase		
3.1	$120\% = 72$ $\text{original} = 72 \div 1.2 = 60$	**60 KD**
3.2	$110\% = 55$ $\text{original} = 55 \div 1.1 = 50$	**50 KD**
3.3	$130\% = 130$ $\text{original} = 130 \div 1.3 = 100$	**100 KD**
3.4	$105\% = 84$ $\text{original} = 84 \div 1.05 = 80$	**80 KD**
3.5	$125\% = 150$ $\text{original} = 150 \div 1.25 = 120$	**120 KD**
3.6	$115\% = 46$ $\text{original} = 46 \div 1.15 = 40$	**40 KD**
price before a reduction		
3.7	$75\% = 36$ $\text{original} = 36 \div 0.75 = 48$	**48 KD**
3.8	$65\% = 52$ $\text{original} = 52 \div 0.65 = 80$	**80 KD**

3.9	$90\% = 63$ $\text{original} = 63 \div 0.9 = 70$	**70 KD**
3.10	$90\% = 45$ $\text{original} = 45 \div 0.9 = 50$	**50 KD**
3.11	$80\% = 56$ $\text{original} = 56 \div 0.8 = 70$	**70 KD**
3.12	$60\% = 120$ $\text{original} = 120 \div 0.6 = 200$	**200 KD**